

Homework 16

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$$1 \quad (Q \vee R) \implies P \equiv (Q \implies P) \wedge (R \implies P)$$

$$(Q \vee R) \implies P \equiv (Q \implies P) \wedge (R \implies P) \quad (1)$$

$$\neg(Q \vee R) \vee P \equiv (Q \implies P) \wedge (R \implies P), \text{Material Implication} \quad (2)$$

$$(\neg Q \wedge \neg R) \vee P \equiv (Q \implies P) \wedge (R \implies P), \text{De Morgan} \quad (3)$$

$$(\neg Q \vee P) \wedge (\neg R \vee P) \equiv (Q \implies P) \wedge (R \implies P), \text{Distribution} \quad (4)$$

$$(Q \implies P) \wedge (R \implies P) \equiv (Q \implies P) \wedge (R \implies P), \text{Material Implication} \quad (5)$$

$$\therefore (Q \vee R) \implies P \equiv (Q \implies P) \wedge (R \implies P)$$

2 $((p \vee q) \wedge \neg p) \implies q$ เป็น Tautology

$$((p \vee q) \wedge \neg p) \implies q \equiv T \quad (6)$$

$$\neg((p \vee q) \wedge \neg p) \vee q \equiv T, \text{Material Implication} \quad (7)$$

$$(\neg(p \vee q) \vee p) \vee q \equiv T, \text{DeMorgan} \quad (8)$$

$$\neg(p \vee q) \vee (p \vee q) \equiv T, \text{Association} \quad (9)$$

$$\therefore ((p \vee q) \wedge \neg p) \implies q \text{ เป็น Tautology}$$